Confidence Interval for Difference in Two Means

Presentation Outline

Variables and Data

Variables

- One Quantitative Variable
- One Categorical Variable with 2 categories indicating sample/population membership

Data Collection

- One Sample containing Two Populations - Divide sample into two groups based on population membership
- Two Samples - one from each Population

Compare observations of quantitative variable between two populations

Parameters for Quantitative Variable

Population 1

- Mean: μ_1
- Variance: σ_1^2
- Std. Dev.: σ_1

Population 2

- Mean: μ_2
- Variance: σ_2^2
- Std. Dev.: σ_2

Data Table

Quantitative Variable	Categorical Variable
Y_{11}	Population 1
Y_{12}	Population 1
$\vdots$	$\vdots$
$\vdots$	$\vdots$
Y_{1n_1}	Population 1
Y_{21}	Population 2
Y_{22}	Population 2
$\vdots$	$\vdots$
$\vdots$	$\vdots$
Y_{2n_2}	Population 2

Summary Statistics

Sample 1

- Mean: $\overline{Y}_1 = \frac{\sum_{j=1}^{n_1} Y_{1j}}{n_1}$
- Variance: $S_1^2 = \frac{\sum_{j=1}^{n_1}(Y_{1j}-\overline{Y}_1)^2}{n_1-1}$
- Std. Dev.: $S_1 = \sqrt{\frac{\sum_{j=1}^{n_1}(Y_{1j}-\overline{Y}_1)^2}{n_1-1}}$

Sample 2

- Mean: $\overline{Y}_2 = \frac{\sum_{j=1}^{n_2} Y_{2j}}{n_2}$
- Variance: $S_2^2 = \frac{\sum_{j=1}^{n_2}(Y_{2j}-\overline{Y}_2)^2}{n_2-1}$
- Std. Dev.: $S_2 = \sqrt{\frac{\sum_{j=1}^{n_2}(Y_{2j}-\overline{Y}_2)^2}{n_2-1}}$

Inference for $\mu_1 - \mu_2$

Goal: Obtain an interval of reasonable values of $\mu_1 - \mu_2$ given the data.

Estimate $\mu_1 - \mu_2$ with $\overline{Y}_1 - \overline{Y}_2$

What is the sampling distribution of $\overline{Y}_1 - \overline{Y}_2$?

Sampling Distribution of $\overline{Y}_1 - \overline{Y}_2$

For large sample sizes n_1 and n_2, we have:

$$\frac{\overline{Y}_1 - \overline{Y}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a standard Normal distribution.

In order to use above result for inference, we would need to know σ_1^2 and σ_2^2.

σ_1^2 and σ_2^2 are population parameters and are generally unknown.

Equal Variance Assumption

Assume $\sigma_1^2 = \sigma_2^2 = \sigma^2$

With this assumption, for large sample sizes n_1 and n_2,

$$\frac{\overline{Y}_1 - \overline{Y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a standard Normal distribution.

Pooled Sample Variance

Estimate the unknown parameter σ^2 with S_p^2.

$S_p^2 =$ pooled sample variance

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{\sum_{j=1}^{n_1}(Y_{1j} - \overline{Y}_1)^2 + \sum_{j=1}^{n_2}(Y_{2j} - \overline{Y}_2)^2}{n_1 + n_2 - 2}$$

Sampling Distribution of $\overline{Y}_1 - \overline{Y}_2$

Replacing σ with S_p gives

$$t = \frac{\overline{Y}_1 - \overline{Y}_2 - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

t has a t distribution with $n_1 + n_2 - 2$ degrees of freedom for large sample sizes n_1 and n_2.

Confidence Interval for $\mu_1 - \mu_2$

For large sample sizes n_1 and n_2, we have:

$$P\left(-t_{\alpha/2} < \frac{\overline{Y}_1 - \overline{Y}_2 - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} < t_{\alpha/2}\right) = 1 - \alpha$$

where $t_{\alpha/2}$ is the upper $\alpha/2$ percentile from a t distribution with $n_1 + n_2 - 2$ degrees of freedom.

Solving the above inequality for $\mu_1 - \mu_2$ gives us:

Confidence Interval for $\mu_1 - \mu_2$

$$P\left(\overline{Y}_1 - \overline{Y}_2 - t_{\alpha/2} * S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < \overline{Y}_1 - \overline{Y}_2 + t_{\alpha/2} * S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}\right) = 1 - \alpha$$

$(1 - \alpha)$% Confidence Interval for $\mu_1 - \mu_2$

$$\overline{Y}_1 - \overline{Y}_2 \pm t_{\alpha/2} * S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Example - Confidence Interval for $\mu_1 - \mu_2$

Creativity Data Sample Statistics

Group	Mean	Std. Dev.	Sample Size
Extrinsic	15.74	5.253	23
Intrinsic	19.88	4.440	24

$$S_p^2 = \frac{(n_1 - 1) * S_1^2 + (n_2 - 1) * S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{22 * 5.253^2 + (23) * 4.440^2}{45}$$

$$= 23.562$$

Example - Confidence Interval for $\mu_1 - \mu_2$

95% Confidence Interval for $\mu_1 - \mu_2$

$$\overline{Y}_1 - \overline{Y}_2 \pm t_{\alpha/2} * S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 15.74 - 19.88 \pm 2.014 * \sqrt{23.562}\sqrt{\frac{1}{23} + \frac{1}{24}}$$

$$= -4.14 \pm 2.853 = (-6.997, -1.291)$$

Example - Confidence Interval for $\mu_1 - \mu_2$

We are 95% confident the mean creativity score of the Extrinsic group is between 1.291 to 6.997 points lower than the mean creativity score of the Intrinsic group.